

Customer Churn Prediction Using Data Mining Techniques: A Comparative Study of Classification Algorithms

Abstract

Customer churn is a major challenge for telecommunication companies because losing existing customers can reduce revenue and increase customer acquisition costs. Data mining and machine learning techniques can help telecom providers identify customers who are likely to discontinue their services and support proactive retention strategies. This study presents a comparative analysis of five classification algorithms for customer churn prediction: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost). The IBM Telco Customer Churn dataset containing 7,043 customer records was used. Data preprocessing included removal of the customer identifier, conversion of the TotalCharges attribute to numeric form, missing-value treatment, categorical-variable encoding, and stratified division into 80% training and 20% testing data. Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Logistic Regression achieved the best overall performance with 80.70% accuracy, 56.68% recall, 60.92% F1-score, and 84.16% ROC-AUC. XGBoost achieved the highest precision of 65.89%. The results indicate that a relatively simple linear model can outperform more complex algorithms on this dataset when appropriate preprocessing is applied.

Keywords: Customer churn, data mining, machine learning, classification, telecom, Logistic Regression, Random Forest, SVM, XGBoost.

I. INTRODUCTION

Customer churn refers to the situation in which an existing customer stops using a company's products or services. In the telecommunication industry, high competition and the availability of alternative service providers make customer retention particularly important. Data mining techniques provide organizations with the ability to discover patterns in customer information and predict potential churn before it occurs [1]–[4].

Traditional churn prediction research has investigated statistical methods, decision trees, neural networks, ensemble learning, and other machine learning techniques [2], [5]. More recent studies have emphasized the importance of feature engineering, class imbalance, model interpretability, and explainable artificial intelligence (XAI) [6]–[10]. Recent systematic reviews also show continued interest in combining classical machine learning, ensemble methods, deep learning, and explainability for telecom churn prediction [11]–[14].

Several studies have demonstrated that no single classification algorithm consistently performs best across all churn datasets. For example, ensemble and boosting approaches can capture nonlinear relationships, while Logistic Regression can provide strong and interpretable results on appropriately encoded tabular data [7], [15]. A recent study by Omari *et al.* compared several models using the IBM Telco Customer Churn dataset and reported strong performance from SVM [16]. Similarly, recent research has investigated explainable ensemble models and demonstrated the value of combining prediction accuracy with interpretability [17].

The objective of this study is therefore to **compare the performance of commonly used data mining classification algorithms for telecom customer churn prediction** using a common dataset and consistent preprocessing and evaluation procedures.

II. DATASET AND METHODOLOGY

A. Dataset

The study uses the IBM Telco Customer Churn dataset. The dataset contains **7,043 customer records and 21 attributes**, including demographic characteristics, subscribed services, contract information, payment method, monthly charges, total charges, and churn status. The target variable is Churn, with two classes: Yes and No.

In the dataset, 1,869 customers were classified as churned and 5,174 as non-churned. Thus, approximately 26.5% of customers belong to the churn class.

B. Data Preprocessing

The following preprocessing operations were performed:

1. customerID was removed because it is an identifier rather than a predictive attribute.
2. TotalCharges was converted from text to numeric format.
3. Invalid or missing numeric values were handled using median imputation.
4. Categorical attributes were converted into numerical representations using one-hot encoding.
5. The target variable was converted into binary form, where No = 0 and Yes = 1.
6. The dataset was divided into **80% training and 20% testing sets** using stratified sampling.
7. Feature scaling was applied to Logistic Regression and SVM using a preprocessing pipeline.

C. Classification Algorithms

Five supervised classification algorithms were evaluated.

Logistic Regression: A linear classification method that estimates the probability of a customer belonging to the churn class. It is computationally efficient and relatively interpretable.

Decision Tree: A tree-based model that recursively divides observations according to feature values. Decision trees can capture nonlinear relationships but may overfit training data [18].

Random Forest: An ensemble of multiple decision trees. Random Forest generally provides greater robustness than a single decision tree and can estimate feature importance [19].

Support Vector Machine: SVM constructs an optimal decision boundary between classes and can model nonlinear relationships through kernel functions [20].

XGBoost: An efficient gradient-boosting algorithm that sequentially builds decision trees to minimize prediction error [21]. XGBoost has been widely used for structured/tabular prediction problems.

D. Evaluation Metrics

The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Accuracy measures the proportion of correctly classified customers. Precision indicates the proportion of predicted churners who actually churned. Recall measures the proportion of actual churners correctly identified. F1-score provides a balance between precision and recall. ROC-AUC measures the model's ability to distinguish between churn and non-churn customers [22].

III. EXPERIMENTAL RESULTS AND DISCUSSION

The experiments were implemented using Python with Pandas, NumPy, Scikit-learn, XGBoost, and Matplotlib. The same training/testing strategy and evaluation metrics were applied to all five algorithms.

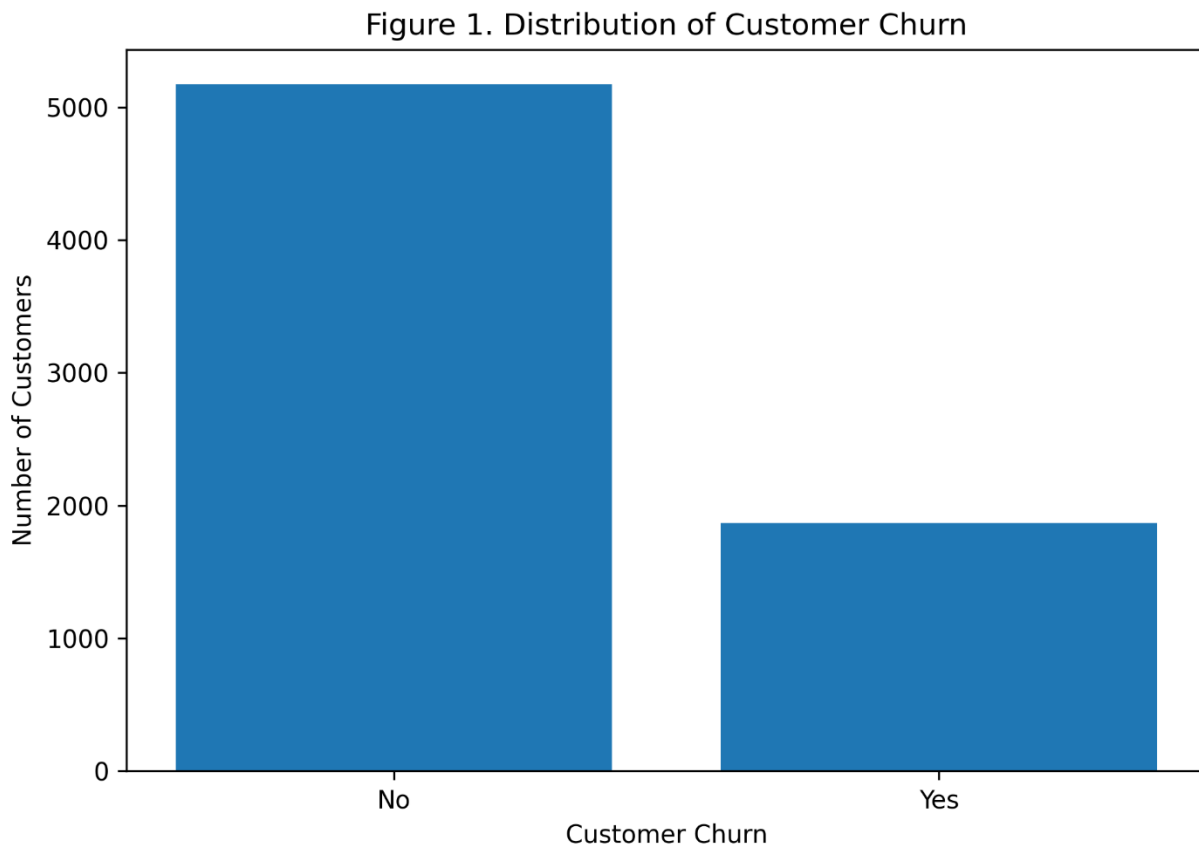
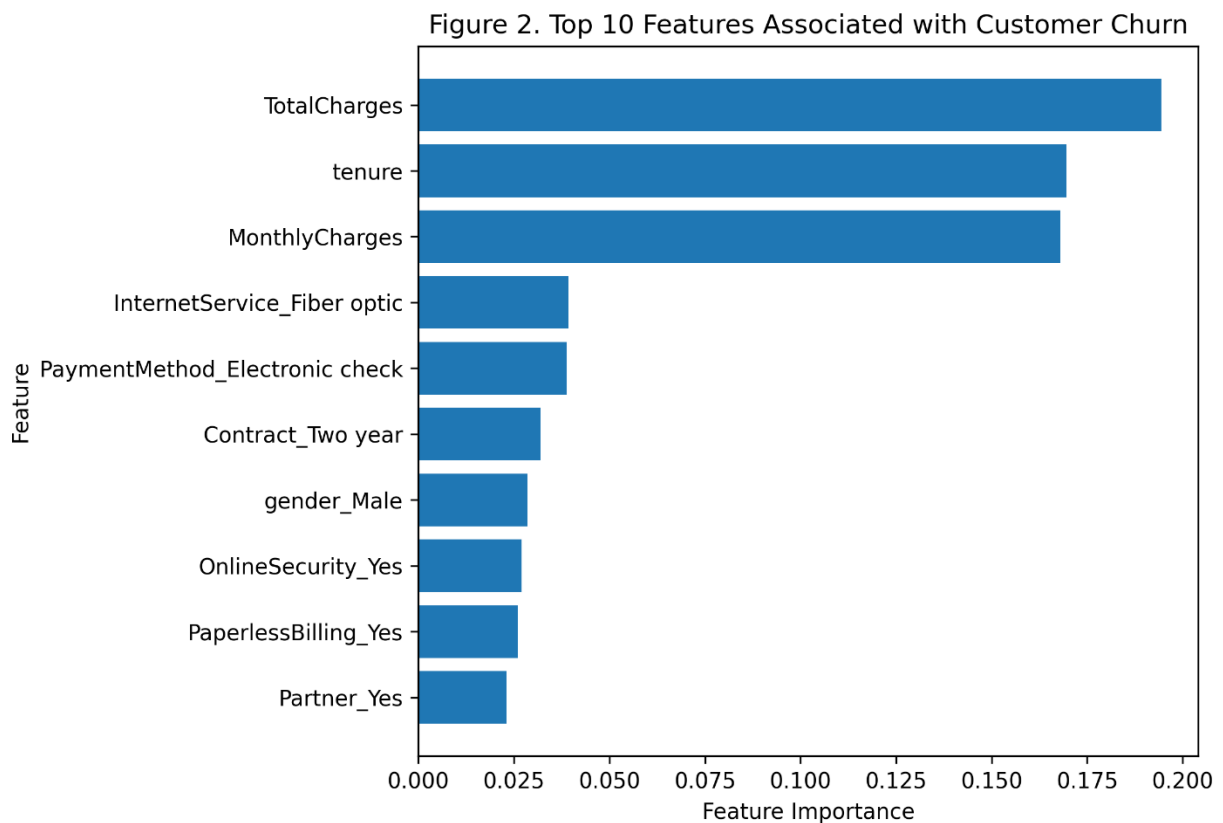


TABLE I

PERFORMANCE COMPARISON OF CLASSIFICATION MODELS

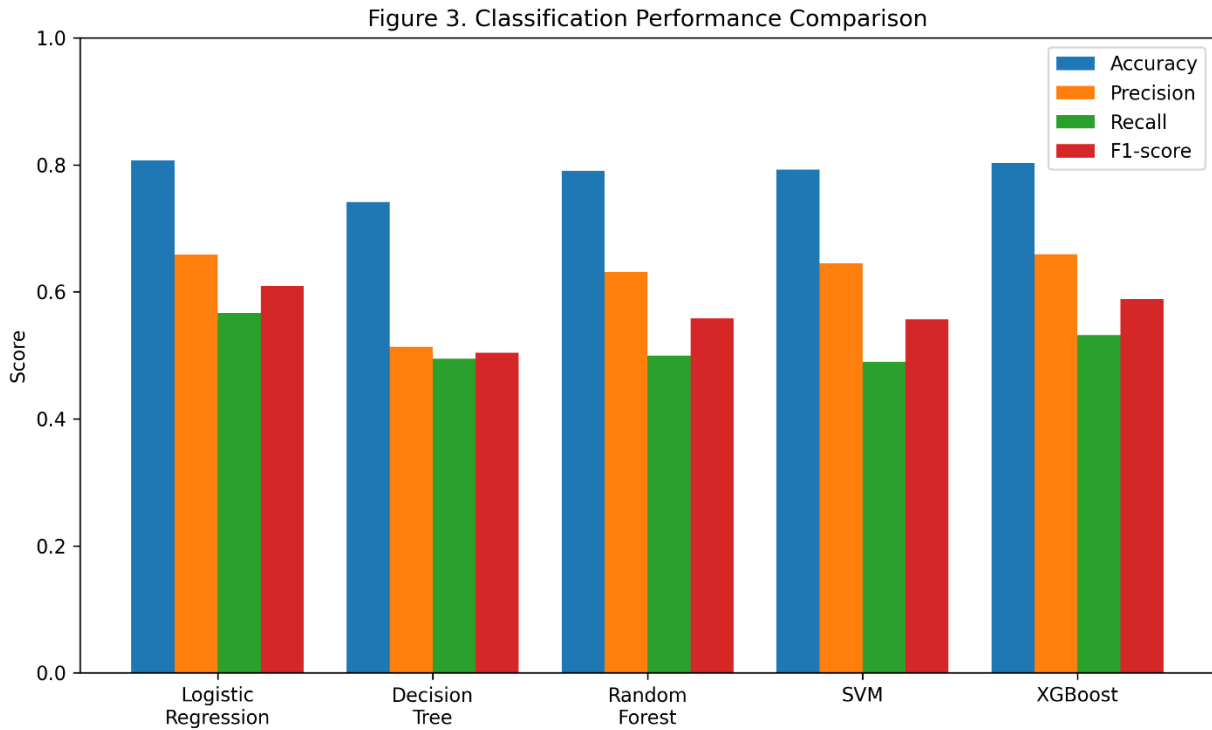
Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	80.70%	65.84%	56.68%	60.92%	84.16%
Decision Tree	74.17%	51.39%	49.47%	50.41%	66.23%
Random Forest	78.99%	63.18%	50.00%	55.82%	82.70%
SVM	79.28%	64.44%	48.93%	55.62%	79.61%
XGBoost	80.27%	65.89%	53.21%	58.88%	84.15%

The results show that **Logistic Regression achieved the strongest overall performance** under the experimental configuration. It obtained 80.70% accuracy and the highest recall, F1-score, and ROC-AUC. Its ROC-AUC of 84.16% indicates good discriminatory capability between churn and non-churn customers.



XGBoost produced almost the same accuracy (80.27%) and ROC-AUC (84.15%) as Logistic Regression and achieved the highest precision of 65.89%. This suggests that XGBoost was effective in identifying customers predicted as churners while reducing false-positive predictions. However, its recall and F1-score were lower than those of Logistic Regression.

Random Forest achieved 78.99% accuracy and 82.70% ROC-AUC. Its performance was better than a single Decision Tree, demonstrating the benefit of ensemble learning. SVM achieved 79.28% accuracy but produced lower recall and F1-score than Logistic Regression.



The Decision Tree showed the weakest overall performance, with 74.17% accuracy and 66.23% ROC-AUC. This result suggests that a single tree was less capable of generalizing the patterns in the dataset compared with the ensemble and linear approaches.

Feature-importance analysis using Random Forest indicated that **TotalCharges**, **tenure**, and **MonthlyCharges** were among the influential variables. This is consistent with the general observation in telecom churn research that customer tenure, service characteristics, contract conditions, and financial variables can provide useful information for predicting customer attrition [6], [7], [15].

The results also demonstrate that greater model complexity does not automatically produce better predictive performance. Although XGBoost and Random Forest can represent complex nonlinear relationships, Logistic Regression produced the best overall results in this experiment. Therefore, model selection should be based on empirical evaluation rather than algorithmic complexity alone.

IV. CONTRIBUTION AND NOVELTY

The main contribution of this study is a controlled comparison of five commonly used classification algorithms using the same telecom churn dataset, preprocessing strategy, train-test split, and evaluation metrics. The study provides a reproducible baseline for comparing linear, tree-based, ensemble, kernel-based, and boosting approaches.

The experimental finding that Logistic Regression outperformed more complex models is also practically relevant. Telecom companies do not necessarily need the most computationally complex algorithm to

obtain useful churn predictions. A model with competitive predictive performance and relatively straightforward interpretation can be advantageous when predictions need to be integrated into customer-retention processes.

The study also combines predictive evaluation with feature-importance analysis, providing an additional perspective on the factors associated with churn. Recent research increasingly emphasizes interpretable and explainable churn prediction rather than accuracy alone [17], [23].

V. CONCLUSION

This study compared Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost for customer churn prediction using the IBM Telco Customer Churn dataset. After consistent preprocessing and evaluation, Logistic Regression achieved the highest accuracy (80.70%), recall (56.68%), F1-score (60.92%), and ROC-AUC (84.16%). XGBoost achieved the highest precision (65.89%) and performed very close to Logistic Regression in accuracy and ROC-AUC.

The findings demonstrate that Logistic Regression was the most effective model under the conditions of this experiment, while Decision Tree produced the weakest results. Important variables identified through Random Forest included TotalCharges, tenure, and MonthlyCharges.

The study is limited by the use of a single publicly available dataset and a single train-test split. Future research could investigate cross-validation, hyperparameter optimization, class-imbalance techniques such as SMOTE, cost-sensitive learning, SHAP/LIME-based explanations, and validation on real telecom datasets from Nepal. Recent research also suggests that explainable ensemble models and CRM integration are promising directions for operational churn management [17], [24].

REFERENCES

- [1] T. Rygielski, J. Wang, and D. Yen, "Data mining techniques for customer relationship management," *Technology in Society*, vol. 24, no. 4, pp. 483–502, 2002.
- [2] Y. Huang, M. A. Kechadi, and D. Buckley, "Customer churn prediction in telecommunications," *Expert Systems with Applications*, vol. 39, no. 1, pp. 1414–1425, 2012.
- [3] S.-H. Hung, D. Yen, and H. Wang, "Applying data mining to telecom churn management," *Expert Systems with Applications*, vol. 31, no. 3, pp. 515–524, 2006.
- [4] A. Nadeem, S. Umar, and M. S. Shahzad, "A review on customer churn prediction data mining modeling techniques," *Indian Journal of Science and Technology*, vol. 11, no. 27, pp. 1–7, 2018.
- [5] A. N. Amin *et al.*, "Cross-company customer churn prediction in telecommunication: A comparison of data transformation methods," *International Journal of Information Management*, vol. 46, pp. 304–319, 2019.
- [6] M. Sana, M. M. Abedin, M. S. Rahman, and M. M. Rahman, "A novel customer churn prediction model for the telecommunication industry using data transformation methods and feature selection," *PLOS ONE*, vol. 17, no. 12, e0278095, 2022.
- [7] U. Lalwani, M. Mishra, and D. Chadha, "Customer churn prediction system: A machine learning approach," *Computing*, vol. 104, pp. 271–294, 2022.

- [8] A. N. Amin *et al.*, "An adaptive learning approach for customer churn prediction in the telecommunication industry using evolutionary computation and Naïve Bayes," *Applied Soft Computing*, vol. 137, p. 110103, 2023.
- [9] M. Barsotti *et al.*, "A decade of churn prediction techniques in the TelCo domain: A survey," *SN Computer Science*, vol. 5, no. 4, p. 404, 2024.
- [10] P. Sikri *et al.*, "Enhancing customer retention in telecom industry with machine learning driven churn prediction," *Scientific Reports*, vol. 14, p. 13097, 2024.
- [11] P. Sharma Poudel, S. Pokharel, and S. Timilsina, "Explaining customer churn prediction in telecom industry using tabular machine learning models," *Machine Learning with Applications*, vol. 17, p. 100567, 2024.
- [12] R. Alotaibi and M. A. Haq, "Customer churn prediction for telecommunication companies using machine learning and ensemble methods," *Engineering, Technology & Applied Science Research*, vol. 14, no. 3, pp. 14572–14578, 2024.
- [13] J. Chang, C. Hall, Y. Xu, A. Amao, S. Ganatra, and T. Benson, "Prediction of customer churn behavior in the telecommunication industry using machine learning models," *Algorithms*, vol. 17, no. 6, p. 231, 2024.
- [14] A. Tebu and B. Izang, "Customer churn prediction in the telecommunication industry over the last decade: A systematic review," *Asian Journal of Research in Computer Science*, vol. 18, no. 4, pp. 256–271, 2025.
- [15] J. Imani, G. Joudaki, S. Beikmohammadi, and H. R. Arabnia, "Customer churn prediction: A systematic review of recent advances, trends, and challenges in machine learning and deep learning," *Machine Learning and Knowledge Extraction*, vol. 7, no. 3, p. 105, 2025.
- [16] A. Omari, O. Al-Omari, T. Al-Omari, and S. M. Fati, "A predictive analytics approach to improve telecom's customer retention," *Frontiers in Artificial Intelligence*, vol. 8, p. 1600357, 2025.
- [17] M. G. Abdelhady and K. A. Mohamed, "Leveraging artificial intelligence for predictive customer churn modeling in telecommunications: A framework for enhanced customer relationship management," *Scientific Reports*, vol. 15, p. 43826, 2025.
- [18] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.
- [19] A. Criminisi and J. Shotton, *Decision Forests for Computer Vision and Medical Image Analysis*. London, U.K.: Springer, 2013.
- [20] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, pp. 273–297, 1995.
- [21] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [22] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.

- [23] M. T. Ribeiro, S. Singh, and C. Guestrin, “‘Why should I trust you?’: Explaining the predictions of any classifier,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.
- [24] S. Arockia Panimalar, A. Krishnakumar, and S. Senthil Kumar, “Intensified customer churn prediction: Connectivity with weighted multi-layer perceptron and enhanced multipath back propagation,” *Expert Systems with Applications*, vol. 265, p. 125993, 2025.
- [25] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [26] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [27] S. Mettle, “Machine learning-based customer churn prediction in telecommunication industry,” *Applied Computational Intelligence and Soft Computing*, 2026, Art. no. 8496186.
- [28] M. Jain, P. Khunteta, and A. Sharma, “Telecom customer churn prediction: A detailed systematic review of data, methodology, and evaluation methods,” *Telecommunication Systems*, vol. 89, no. 3, 2026.
- [29] A. Hooda *et al.*, “Combining predictive accuracy and interpretability: A data-driven approach to telecom churn analysis,” *Scientific Reports*, vol. 16, p. 4596, 2026.